

Table 64. IWDG min/max timeout period at 32 kHz (LSI)

For the values in this table, the exact timings still depend on the phasing of the APB interface clock versus the LSI clock, so that there is always a full RC period of uncertainty.

Prescaler divider	PR[2:0] bits	Min timeout RL[11:0] = 0x000	Max timeout RL[11:0] = 0xFF	Unit
/4	0	0.125	512	ms
/8	1	0.250	1024	
/16	2	0.500	2048	
/32	3	1.0	4096	
/64	4	2.0	8192	
/128	5	4.0	16384	
/256	6 or 7	8.0	32768	

5.3.24 I3C interface characteristics

The I3C interface meets the timing requirements of the MIPI® I3C specification v1.1.

The I3C peripheral supports:

- I3C SDR-only as controller
- I3C SDR-only as target
- I3C SCL bus clock frequency up to 12.5 MHz

The parameters given in Table 65 below are obtained with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- I/O Compensation cell activated
- Voltage scaling range 1

The I3C timings are in line with the MIPI specification, except for the ones given in Table 65, I3C open-drain measured timing. For t_{SU_OD} , this can be mitigated by increasing the corresponding SCL low duration in the I3C_TIMINGR0 register. For further details refer to AN5879.

Table 65. Open drain timing measurements

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Conditions	Min	Unit
t_{SU_OD}	SDA data setup time in open drain mode	Controller $2.7\text{ V} \leq V_{DDIOX} \leq 3.6\text{ V}$	24.5 ⁽¹⁾	ns

1. The minimum SDA data setup time during open-drain-mode is 3 ns, as specified in the MIPI Alliance specification for I3C.

5.3.25 I2C interface characteristics

The I2C interface meets the timing requirements of the I2C-bus specification and user manual rev. 03 for:

- Standard mode (Sm): Bit rate up to 100 kbit/s.
- Fast mode (Fm): Bit rate up to 400 kbit/s.
- Fast mode plus (Fm+): Bit rate up to 1 Mbit/s.

The I2C timing requirements are specified by design, not tested in production, when the I2C peripheral is properly configured (refer to the product reference manual).

The SDA and SCL I/O requirements are met with the following restrictions:

- The SDA and SCL I/O pins are not true open-drain. When configured as open-drain, the PMOS connected between the I/O pin and V_{DDIOX} is disabled but remains present.
- Only FT_f I/O pins support Fm+ low level output current maximum requirement. Refer to Section 5.3.14: I/O port characteristics for the I2C I/Os characteristics.

All I2C SDA and SCL I/Os embed an analog filter. Refer to the table below for the analog filter characteristics.